Physics 3221 Spring 2026: Midterm 2

Grade:

1. A Strange Force (20pts)

A particle travels between points A and B as shown in Fig. 1. The particle is subject to the following force:

$$\vec{F} = k\,y\hat{x} + l\frac{dy}{dx}\hat{y}\,,$$

where k and l are constants.

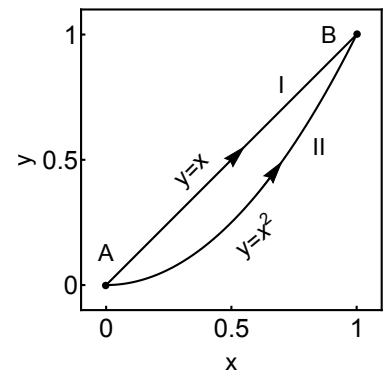

Figure 1: Points A and B connected by various paths

(a) **[5 pts]** When $l = 0$, is this force path-independent? Verify by calculating its curl $\vec{\nabla} \times \vec{F}$.

(b) **[10 pts]** Calculate the work $\vec{F}$ does on the particle if it travels between point A and point B along path I given by $y = x$

(c) **[5 pts]** Calculate the work $\vec{F}$ does on the particle if it travels between point A and point B along path II given by $y = x^2$

Grade:

2. The Damped Harmonic Oscillator (20 Points)

A mass attached to a spring as depicted in Fig. 2 experiences only the following forces:

$$\text{Spring force:} \quad \vec{F}_{\text{s}} = -k\vec{x}$$

$$\text{Drag force:} \quad \vec{F}_d = -b\vec{v}$$

where $\vec{x}$ is defined such that at $\vec{x} = 0$, the spring is at its equilibrium position. All motion of the spring occurs along the x-axis.

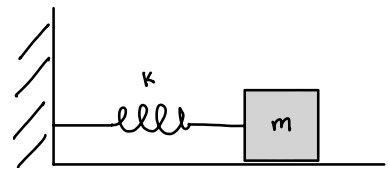

Figure 2: A mass m attached to a spring with spring constant k, moving only in the x-direction

(a) **[7 pts]** Use Newton's 2nd Law to write down a differential equation describing the motion of the spring, and solve it to find the general solution $x(t)$ for the spring. *Note: I have not yet given you initial conditions, so you can leave two undetermined constants in your answer.*

(b) **[6 pts]** If $k = b^2/m$, is the system oscillating? Why or why?

THE REST OF PROBLEM 2 IS ON THE NEXT PAGE

(c) **[7 pts]** Assume $k = b^2/m$. At $t = 0$, assume the mass is released from rest at position x_0. At time t_1, the mass is at a local maximum and turning back around towards equilibrium, where t_1 is given by:

$$t_1 = 2\pi/\omega_1 \,, \qquad \omega_1 = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \,.$$

How much work has been done by $\vec{F}_d$ from $t = 0$ to $t = t_1$? You will get full credit for obtaining the correct magnitude of work (regardless of sign).

Grade:

3. Driven Harmonic Oscillator (20 Points)

Consider now the mass attached to a spring as shown in Fig. 2 being acted on by a periodic driving force $F(t)$. A plot for $f(t)$ is given in Fig. 3, where $f(t) \equiv F(t)/m$. Below, fill in the blanks for the following derivation for the solution $x(t)$ that gives the resulting motion of the mass.

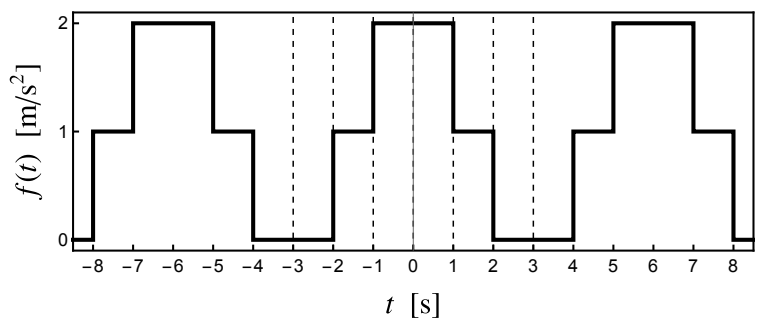

Figure 3: The function $f(t)$ related to the driving force $F(t)$. As shown in the plot, $f(t)$ has a period of 6 s.

The equation of motion for this mass on a spring is:

$$\ddot{x}(t) + 2\beta\dot{x}(t) + \omega_0^2 x(t) = f(t)$$

(a) **[3 pts]** Find β and ω_0.

Next, use the Fourier expansion for $f(t)$:

$$f(t) = \sum_n a_n \cos(n\omega t) + b_n \sin(n\omega t)$$

(b) **[4 pts]** Find a_0

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(c) **[4 pts]** Find a_n for $n > 0$.

(d) **[4 pts]** Find b_n for general n.

(e) **[1 pt]** Given a_0, a_n $(n > 0)$, and b_n, write $f(t)$ as a Fourier expansion:

$$f(t) = \underline{\hspace{6cm}} + \sum_{n=1}^{\infty} \underline{\hspace{8cm}}$$

Now we have $f(t)$ as a sum of $f_n(t)$ functions, so we can write the general solution to the equation of motion $x(t)$ as:

$$f(t) = \sum_{n=0}^{\infty} f_n(t) \qquad \Rightarrow \qquad x(t) = \sum_{n=0}^{\infty} x_n(t)\,,$$

where each $x_n(t)$ solves the following differential equation:

$$\ddot{x}_n(t) + 2\beta\dot{x}_n(t) + \omega_0^2 x_n(t) = f_n(t)$$

We also know the following differential equations have the following solutions:

$$\ddot{y}_1(t) + 2\beta\dot{y}_1(t) + \omega_0^2 y_2(t) = f_0 \cos\omega t \qquad \Rightarrow \qquad y_1(t) = A\cos(\omega t - \delta)$$
$$\ddot{y}_2(t) + 2\beta\dot{y}_2(t) + \omega_0^2 y_2(t) = f_0 \sin\omega t \qquad \Rightarrow \qquad y_2(t) = A\sin(\omega t - \delta)$$

where A and δ are on the equation sheet.

(f) **[3 pts]** Using the information above, find $x_n(t)$

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(g) **[1 pt]** Add the x_n solutions together to get the full solution from the motion:

$$x(t) = \sum_{n=0}^{\infty} x_n(t) = \ \mathbf{?}$$

Formulas (just for the Midterm)

Newton's 2nd Law

$$\vec{F} = m\ddot{\vec{r}}$$

Cartesian

$$F_x = m\ddot{x} \qquad F_y = m\ddot{y} \qquad F_z = m\ddot{z}$$

Polar

$$F_r = m(\ddot{r} - r\dot{\phi}^2) \qquad F_\phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi})$$

Cylindrical

$$F_\rho = m(\ddot{\rho} - \rho\dot{\phi}^2) \quad F_\phi = m(\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}) \quad F_z = m\ddot{z}$$

Coordinate Systems

Cartesian: $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$d\vec{r} = dx\hat{x} + dy\hat{y} + dz\hat{z}$

Cylindrical : $\vec{r} = \rho\hat{\rho} + z\hat{z}$

$d\vec{r} = d\rho\hat{\rho} + \rho d\phi\hat{\phi} + dz\hat{z}$

Spherical: $\vec{r} = r\hat{r}$

$d\vec{r} = dr\hat{r} + rd\theta\,\hat{\theta} + r^2\sin\theta\, d\phi\,\hat{\phi}$

Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Fourier Series

$$f(t) = \sum_{n=0}^{\infty}\left[a_n\cos(n\omega t) + b_n\sin(n\omega t)\right],$$

$$\omega = 2\pi/\tau.$$

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,,$$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos(n\omega t)dt\,, \qquad n > 0$$

$$b_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\sin(n\omega t)dt\,,$$

Hook's Law & Simple Harmonic Motion

$$F = -kx \Leftrightarrow U = \frac{1}{2}kx^2$$

$$x(t) = A\cos(\omega t - \delta)$$

Driven Oscillator

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2)}$$

$$\delta = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right)$$

Work

$$W = \int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,')\cdot d\vec{r}\,'$$

Conservative forces

$$U(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,')\cdot d\vec{r}\,'$$

$$F = -\vec{\nabla}U$$

$$\vec{\nabla}\times\vec{F} = 0$$

Useful Integrals

$$\int x\sin(ax)dx = -\frac{x}{a}\cos(ax) + \frac{1}{a^2}\sin(ax)$$

$$\int x\cos(ax)dx = \frac{x}{a}\sin(ax) + \frac{1}{a^2}\cos(ax)$$

Gradient and Curl

$$\vec{\nabla}f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla}f = \hat{r}\frac{\partial f}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial f}{\partial\theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial f}{\partial\phi}$$

$$\vec{\nabla}f = \hat{\rho}\frac{\partial f}{\partial r\rho} + \hat{\theta}\frac{1}{\rho}\frac{\partial f}{\partial\theta} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla}\times\vec{A} = \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$